

PROGRAM – 7

Deploy a Web Application to Kubernetes using Minikube

Project Name: program_devops_7

Technologies: Kubernetes, Minikube, Docker, Node.js

System Requirements

Hardware Requirements

- **Minimum 4 GB RAM**
- **Minimum 20 GB free disk space**
- **Dual Core Processor or higher**

Software Requirements

- **Ubuntu / Windows / macOS**
- **Docker Engine**
- **Minikube**
- **kubectl**
- **Node.js**

Step 1 – Start Minikube

minikube start

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_7
meghana@meghana-HP-Laptop-15s-dr3xxx:~$ minikube start
minikube v1.37.0 on Ubuntu 24.04
👉 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
👉 Pulling base image v0.0.48 ...
👉 Restarting existing docker container for "minikube" ...
👉 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
👉 Verifying Kubernetes components...
   ▪ Using image docker.io/kubernetes/dashboard:v2.7.0
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

👉 Enabled addons: storage-provisioner, default-storageclass, dashboard
👉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
meghana@meghana-HP-Laptop-15s-dr3xxx:~$ kubectl get nodes
NAME       STATUS   ROLES    AGE   VERSION
minikube   Ready    control-plane  40d   v1.34.0
meghana@meghana-HP-Laptop-15s-dr3xxx:~$ mkdir program_devops_7
meghana@meghana-HP-Laptop-15s-dr3xxx:~$ cd program_devops_7
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ nano app.js
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ nano package.json
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ nano Dockerfile
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$
```

Step 2 – Node.js Application (app.js)

```
GNU nano 7.2 app.js
const express = require("express");
const app = express();

app.get("/", (req, res) => {
  res.send("Hello from Kubernetes using Minikube - program_devops_7");
});

app.get("/health", (req, res) => {
  res.json({ status: "UP" });
});

app.listen(3000, "0.0.0.0", () => {
  console.log("App running on port 3000");
});

```

Wrote 14 lines

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward

Step 3 – Dependency File (package.json)

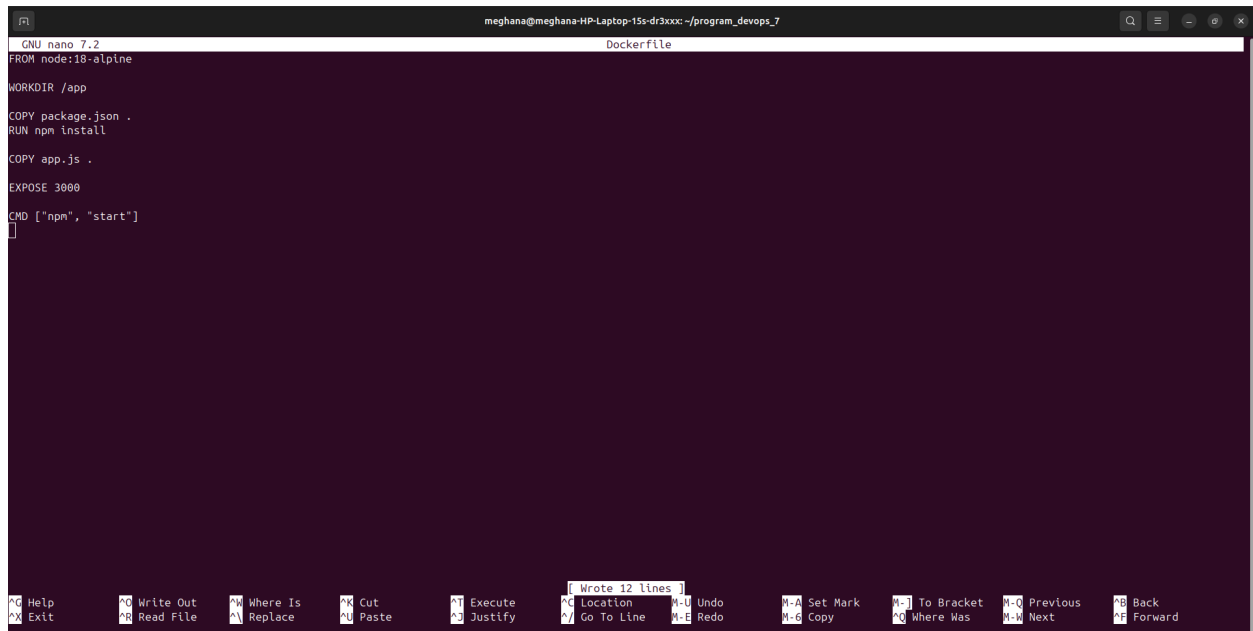


The screenshot shows a terminal window with the nano editor open, editing a file named `package.json`. The editor's title bar indicates the user is `meghana` on a `meghana-HP-Laptop-15s-dr3xxx` in the `-/program_devops_7` directory. The file content is as follows:

```
GNU nano 7.2 package.json
{
  "name": "program_devops_7",
  "version": "1.0.0",
  "main": "app.js",
  "scripts": {
    "start": "node app.js"
  },
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

The bottom status bar of the nano editor shows various keyboard shortcuts for editing and navigation, such as `Ctrl+O` for Help, `Ctrl+X` for Exit, `Ctrl+W` for Write Out, and `Ctrl+S` for Save.

Step 4 – Dockerfile



The screenshot shows a terminal window with the nano editor open, editing a file named `Dockerfile`. The editor's title bar indicates the user is `meghana` on a `meghana-HP-Laptop-15s-dr3xxx` in the `-/program_devops_7` directory. The file content is as follows:

```
GNU nano 7.2 Dockerfile
FROM node:18-alpine

WORKDIR /app

COPY package.json .
RUN npm install

COPY app.js .

EXPOSE 3000

CMD ["npm", "start"]
```

The bottom status bar of the nano editor shows various keyboard shortcuts for editing and navigation, such as `Ctrl+O` for Help, `Ctrl+X` for Exit, `Ctrl+W` for Write Out, and `Ctrl+S` for Save.

Step 5 – Build Docker Image in Minikube

```
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ eval $(minikube docker-env)
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ docker build -t program_devops_7 .
[+] Building 13.7s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 158B
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:18-alpine
=> [internal] load build context
=> => transferring context: 568B
=> [2/5] WORKDIR /app
=> [3/5] COPY package.json .
=> [4/5] RUN npm install
=> [5/5] COPY app.js .
=> exporting to image
=> => exporting layers
=> => writing image sha256:1386d630fcd8bfa60db779faa32d906f760f3aa5c714088141959d7eaf77edc54
=> => naming to docker.io/library/program_devops_7
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$
```

Step 6 – Kubernetes Deployment

```
GNU nano 7.2 deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: program-devops-7-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: program-devops-7
  template:
    metadata:
      labels:
        app: program-devops-7
    spec:
      containers:
        - name: node-app
          image: program_devops_7
          imagePullPolicy: Never
          ports:
            - containerPort: 3000
```

Step 7 – Apply Deployment

kubectl apply -f deployment.yaml

```
=> exporting to image
=> => exporting layers
=> => writing image sha256:1386d630fcd8bfa60db779faa32d906760f3aa5c714088141959d7eaf77edc54
=> => naming to docker.io/library/program_devops_7
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ nano deployment.yaml
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ kubectl apply -f deployment.yaml
deployment.apps/program-devops-7-deployment created
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
demo-app-787c9b7c57-nmffk           1/1     Running   5 (68m ago)  18d
demo-app-787c9b7c57-q7bz4           1/1     Running   5 (68m ago)  18d
my-first-pod                         1/1     Running   1 (68m ago)  143m
nginx-deployment-599786d4bc-6xdvs    1/1     Running   3 (68m ago)  4d21h
nginx-deployment-599786d4bc-chbsf    1/1     Running   3 (68m ago)  4d21h
nginx-pod                            0/1     Completed 0            145m
nodejs-pod                           1/1     Running   5 (3m15s ago)  18d
program-devops-7-deployment-56bf5757cf-2qqqg 1/1     Running   0            4s
program-devops-7-deployment-56bf5757cf-p9cqg 1/1     Running   0            4s
python-hello-7f4448c876-zqdz7        1/1     Running   7 (3m15s ago)  18d
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$
```

Step 9 – Apply Service

kubectl apply -f service.yaml

```
GNU nano 7.2                                service.yaml
apiVersion: v1
kind: Service
metadata:
  name: program-devops-7-service
spec:
  type: NodePort
  selector:
    app: program-devops-7
  ports:
    - port: 3000
      targetPort: 3000
      nodePort: 30007
  
```

Wrote 12 lines

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward

```
program-devops-7-deployment-56bf5757cf-2qqqg 1/1 Running 0 4s
program-devops-7-deployment-56bf5757cf-p9cqp 1/1 Running 0 4s
python-hello-7f4448c876-zqdz7 1/1 Running 7 (3m15s ago) 18d
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ nano service.yaml
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ kubectl apply -f service.yaml
service/program-devops-7-service created
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ minikube service program-devops-7-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	program-devops-7-service	3000	http://192.168.49.2:30007

```
🌐 Opening service default/program-devops-7-service in default browser...
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_7$ [17326:17326:0105/124622.077838:ERROR:chrome/browser/profiles/pr
[17326:17326:0105/124622.077913:ERROR:chrome/browser/profiles/profile_attributes_storage.cc:1032] Failed to PNG encode th
[17326:17326:0105/124622.850565:ERROR:chrome/browser/ui/views/user_education/impl/browser_user_education_interface_impl.c
zation complete; IPH will not be shown.
[17326:17326:0105/124622.897275:ERROR:chrome/browser/ui/views/user_education/impl/browser_user_education_interface_impl.c
zation complete; IPH will not be shown.
```

Step 10 – Access Application

minikube service program-devops-7-service

